

GRANT – LETTER OF INQUIRY FORM

First Name [REDACTED]

Last Name [REDACTED]

Email [REDACTED]

Phone [REDACTED]

Non-Profit Organization Name [REDACTED]

EIN Number [REDACTED]

Amount Requested [REDACTED]

**Please briefly describe your organization and mission:
(Max 100 words)**

The [REDACTED] is a 501(c)3 nonprofit organization focused on identifying the city's climate risks and vulnerabilities, and their impact on public health and safety, our city's economy, and our natural resources. The [REDACTED] includes a member of the city council and representatives of local and regional climate and environmental nonprofits, including the Sierra Club.

More information is available at [REDACTED]

Mission: *Empower [REDACTED] to achieve net-zero greenhouse gas emissions by 2050, strengthen resilience to climate change, and inspire a sustainable and equitable future for all.*

**Briefly provide how this project fits in within your overall organizational budget and any other funding or partnerships secured for the proposed project.
(Max 100 words)**

The [REDACTED] has embarked on a dual-track program that combines fundraising and grant acquisition goals for specific projects focused on climate risks and vulnerabilities. A portion of our overall budget has been used to begin tree mapping, and we have developed an internship program with a local college to help advance this project while also providing education to students in tree mapping technology. Grant funding will also enable us to provide actionable results to the city, which has provided us with more recent high-resolution imagery to assist in this project.

Provide a short description of the project you are proposing and address approach, significance and feasibility. If your project is of a technical nature, provide us with details to help us understand the impact. *

(250 words max)

The *Tree Canopy Mapping Project* applies machine learning that leverages cutting-edge computer vision and Geographic Information Systems analysis to provide an affordable, scalable, and replicable method for counting and mapping individual trees across the city of ████████ – a method that can potentially scale across cities large and small. The project integrates artificial intelligence and high-resolution aerial imagery, adapting methods from recent research on Deep Distance Transformers and applying them to tree canopy imagery.

Outputs include a geospatial shapefile containing all mapped tree instances for ████████, a preliminary canopy count, and distribution analysis. Subsequent deliverables can integrate canopy data with demographic and temperature data to explore equity and heat vulnerability correlations and a digital canopy model for urban heat island analysis.

Resulting datasets will equip city planners with essential canopy data for sustainable development and enable residents and climate action volunteer organizations to understand local environmental conditions and resilience opportunities.

This initiative represents a bridge between science, technology, and civic engagement – a model for how community organizations can use advanced spatial analysis to inform equitable climate action. Without accurate assessments of natural assets like urban forests, cities cannot effectively plan for conservation or restoration. The project bridges that gap by creating a spatially explicit, data-driven foundation for understanding tree canopy distribution, density, and change over time. More broadly, it demonstrates how AI and open-source tools can make advanced environmental analysis accessible to small municipalities and nonprofits. By democratizing resource assessment, the project supports data-driven climate planning and transparent community engagement.

Please describe how this project aligns with and supports our mission. *

(200 words max)

This project leverages technology and science to preserve the myriad benefits of urban forests, supporting Sustainable Path's dedication to regional sustainability and health. Foundational to better management of ████████ tree canopies, this mapping approach is a powerful opportunity to mitigate climate impacts and protect public health by catalyzing optimization of contributions that trees make to a sustainable future:

(1) sequestering carbon; (2) reducing pollution (filtering air and producing oxygen); (3) conserving energy (cooling buildings); (4) improving stormwater management (roots absorb water, reducing runoff and flooding); (5) preserving biodiversity (providing habitat and food for pollinators, birds, other wildlife); (6) making cities more economically sustainable, reducing energy bills and infrastructure costs (shade reduces pavement cracking and roots filter water sources), and increasing property values (homes and shopping areas on tree-lined-streets); (7) improving public health well beyond cleaner air, with evidence-based mental health benefits of the calming effects of nature and encouragement of outdoor activities for physical health.

This project provides the foundational data that systems thinking requires for understanding how trees underpin regional sustainability and public health. It transforms urban forestry from reactive maintenance to proactive management, enabling evidence-based decisions about green infrastructure investments, equitable canopy distribution, and climate adaptation strategies.
